JOAN VENDRELL GALLART

Ph.D. in ML&Optimization at University of California Irvine

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EXPERIENCE

Machine Learning Research Scientist Intern
Google Moonshot

Sept 2026 – present | San Francisco, California

- Developing novel Agentic AI framework for Industry and Operational Research Optimization.

Machine Learning Research Scientist Intern
KavAI

Jun 2026 – Sept 2026 | San Francisco, California

- Developing novel Agentic AI framework for Industry. Specifically: mixing Submodular Theory, Reinforcement Learning, Curiosity and Agentic Testing.

Machine Learning Research Scientist Intern
Los Alamos National Laboratory

Jun 2025 – Jun 2026 | Los Alamos, New Mexico

- Developing novel Agentic AI framework for research: URSA - The Universal Research and Scientific Agent. Specifically: mixing Submodular Theory, Distillation, Fine-tuning and test-time hallucination prevention.

Research Scientist Intern
Los Alamos National Laboratory

Jun 2024 – Aug 2024 | Los Alamos, New Mexico

- Probabilistic methods over graphs for electric power grid optimization on PowerModelsDistribution.jl.

Python Full-stack software developer
011h

Apr 2022 – Jul 2022 | Barcelona, Spain

- Full-stack development of a web-app for a wood construction company.

Optimization Research intern
Cooperative Laboratory-UCI

Sep 2021 – Mar 2022 | Irvine, California

- Balsells Mobility Award: Applying Quantum inspired algorithms to solve motion planning problems in multi-agents systems.

Machine Learning Research intern
Institut de Robòtica Informàtica industrial (IRI CSIC-UPC)

Sep 2019 – Jul 2021 | Barcelona, Spain

- JAE Intro Grant: ML and Computer Vision internship at Perception and Manipulation Group.

EDUCATION

Ph.D. in Machine Learning and Optimization
University of California Irvine

Sept 2022 – On going

Thesis title: Trustworthy Decision-Making via Embodied Submodular Theory

M.Sc. in Machine Learning, Robotics and Control
Universitat Politecnica de Catalunya

Sept 2019 – Abr 2022

B.Sc. in Electric Engineering
Universitat Politecnica de Catalunya

Sept 2015 – June 2019

MOST PROUD OF

- **Holmes fellowship** for navigation optimization
- **Henry Samueli Endowed fellowship** for discrete optimization
- **UCI-LANL fellowship** for power systems optimization
- **Balsells Fellowship** for funding my PhD studies
- **Catalan Young Researcher** national award for my research projects
- **Distinction with honours** for my bachelor's thesis

STRENGTHS

Python | Matlab | JavaScript | React | Machine Learning | AI | PyTorch | TensorFlow | Qiskit | Cirq | Git | Pandas | SQL | Big Data | LangChain

Data Structures | Graphs | Optimization | Quantum Algorithms | Finance

Technology Consultant

Institut Idelfons Cerda

Jan 2019 – Jul 2019 Barcelona, Spain

- Consultant of technology for sustainable urban development of Barcelona area.

PUBLICATIONS

Latest Submissions

- J. V. Gallart, M. Grosskopf, and R. Bent, *Hierarchical distillation and adaptive fine-tuning for resource-constrained large language models within agentic ai systems*, Submitted to ICML 2026.
- M. Grosskopf, N. A. Debardeleben, R. Bent, *et al.*, *URSA: The Universal Research and Scientific Agent*, OpenReview preprint, Submitted to ICML 2026.
- J. Vendrell and S. Kia, *Bounding the price of feasibility: Optimality guarantees for conditioned greedy algorithms in network reconfiguration*, Submitted to CDC 2026.
- J. Vendrell and S. Kia, *Large language models are gestalt learners*, Submitted to NIPS 2026.
- J. Vendrell, S. Kia, M. Grosskopf, and R. Bent, *Chain-based adaptive reconfiguration over lattices for hallucination reduction*, Submitted to JMLR 2026.

Conference Proceedings

- J. V. Gallart, A. Kuhnle, and S. Kia, "Resque greedy: Rewiring sequential greedy for improved submodular maximization," arXiv, 2025. DOI: 10.48550/ARXIV.2505.13670.
- R. B. Joan Vendrell and S. Kia, "Forward: Feasibility oriented random-walk inspired algorithm for radial reconfiguration in distribution networks," in *American Automatic Control Council*, 2025. DOI: 10.48550/arXiv.2410.14080.
- J. Vendrell, "Monte carlo sampling and qaoa over probabilistic submodular models," in *Quantum Algorithms in Pre-Fault Tolerant Hardware Basq-IBM conference*, Poster, 2025.
- J. Vendrell and S. Kia, "Optimality gap of decentralized submodular maximization under probabilistic communication," in *Conference on Decision and Control*, 2024. DOI: 10.48550/arXiv.2409.09979.

Journal Articles

- C. Sun, D. Henry, J. Vendrell, V. Puig, and E. Bonet, "Data-driven approaches for pipe life prognosis in water distribution networks: The barcelona water distribution network case study," *Journal of Hydroinformatics*, vol. 27, no. 6, pp. 1003–1018, Jun. 2025, ISSN: 1465-1734. DOI: 10.2166/hydro.2025.003.
- J. Vendrell and S. Kia, "Quantum-inspired evolutionary algorithm for optimal service-matching task assignment," *Information*, vol. 13, p. 438, 2022.
- D. Henry, C. Sun, J. Vendrell, V. Puig, and E. Bonet, "Pipe life prognosis in water distribution networks using reliable data-based approaches," *2021 5th International Conference on Control and Fault-Tolerant Systems (SysTol)*, pp. 187–192, 2021.

Hard-working | Learning | Research | Fuzzy Logic | Time-serie structures | LangChain | Google ADK

LANGUAGES

English ●●●●●

Spanish ●●●●●

Catalan ●●●●●

Portuguese ●●○○○

REFEREES

Prof. Solmaz Kia
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Prof. Russell Bent
@ Los Alamos National Lab
rbent@lanl.gov

Prof. Ramin Bostanabad
@ University of California Irvine
raminb@uci.edu

Dr. Congcong Sun
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Ricard Gine
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PEER REVIEWER

- American Control Conference
- Control and Decision Conference
- IEEE Transaction on Control of Networked Systems
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- International Conference on Machine Learning (IMCL)
- International Conference on Robotics and Automation (ICRA)
- NeurIPS

Books

- J. Vendrell and C. Sun, *Fuzzy clustering application on failure rate prediction in water distribution networks*. Barcelona: IRI, 2020.
- J. Vendrell and C. A. Bahon, *Genertive Adversarial Neural Networks for High Frequency stocks prediction*. Barcelona: UPC, 2019.

PROJECTS

Physics Simulator

Personal project

On going

- Agentic AI system for physics simulation. PromptPhysicsSim.

GNN Simulator

UCI

2023

- GNN for physics simulation. GitHub.

HoneyAnt

Start-up

On going

- Agentic AI system for consultancy. HoneyAnt.

Academic advisor bot

ETSEIB - UPC

2017

- Processing of academic data with pandas library in Python and implementation of a Web App for smartphones.

Algorithmic Trading: Backtest, Optimize & Automate in Python

Bloom Team

2019

Development of algorithmic trading platform with Python and alert messages/control through Telegram.

ACADEMY

Adjunct Professor - Machine Learning

Universitat Carlemany

On going

- Professor of Machine Learning in the Mathematics Bachelor's of the Universitat Carlemany

Adjunct Professor - Discrete Mathematics

Universitat Carlemany

On going

- Professor of Discrete Mathematics in the Mathematics Bachelor's of the Universitat Carlemany

Teaching Assistant 150 - Solid Mechanics

UCI-MAE

2024

- In charge of 200 students, teaching about tension and stress in materials.

Teaching Assistant 170 - Advance Control

UCI-MAE

2024

- In charge of 200 students, teaching about control and dynamics.

Teaching Assistant 130 - Fluid Mechanics

UCI-MAE

2026

- In charge of 200 students, teaching about fluids mechanics.